Product Requirements Document (PRD)

Hana Hideaway B&B · Hawaii Hospitality Dashboard & Marketing Website

ICS 385 Spring 2026 | University of Hawaii Maui College | Noah Munz

1. Project Overview

Property: Hana Hideaway B&B | **Island:** Maui | **Visitor Segment:** Eco-tourists

Hana Hideaway B&B is a full-stack web application for a locally-owned bed and breakfast in the rainforests of Hana, Maui. The app has three pages: (1) a **Marketing Page** targeting eco-tourists with property info and a booking call-to-action; (2) a **Visitor Statistics Dashboard** showing real Hawaii tourism data from DBEDT with interactive charts; and (3) a password-protected **Admin Panel** for the property owner to post special offers and manage content. The backend uses Node.js, Express, and MongoDB Atlas; the frontend uses React with Vite.

What has been completed so far (Weeks 10–11):

- Week 10: Mongoose Property schema created with 5 seeded Maui B&B records in MongoDB Atlas.
- Week 11: Express server with GET /properties, GET /properties/:id, and POST /properties/:id/reviews routes. EJS template rendering all properties. Postman collection tested and exported.
- Week 12: React marketing page built with Hero, About, Amenities, and CallToAction components. Data fetched live from Express API using useEffect and useState.

2. User Personas

Persona A — Target Visitor (Eco-Tourist): Maya, 31, environmental consultant from Oregon. She browses the marketing page on mobile, looking for sustainable amenities, authentic local experiences, and honest guest reviews before booking.

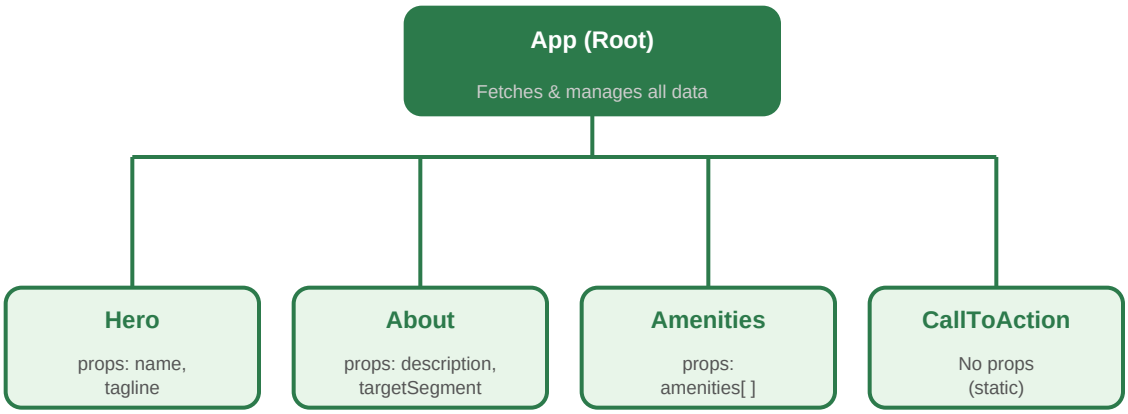
Persona B — Property Admin: Kai, 45, owner of Hana Hideaway B&B. He logs into the admin panel to post seasonal promotions, view occupancy trends from DBEDT data, and manage property listings without writing code.

3. Marketing Page Components Table

Component	Data Displayed	Data Source	User Interaction
Hero	Property name, tagline, beach photo	xpress API + Unsplash image	Click 'Book Your Stay' → scrolls to CTA
About	Description, target visitor segment	xpress	Read-only
Amenities	Amenity list (Ocean View, Hiking, Farm-to-Table)	MongoDB amenities array via API	Responsive grid display
CallToAction	Booking prompt + contact email	Static / hardcoded	Click opens email client

4. Component Hierarchy Diagram

App is the root component and sole data-fetcher; it passes data down to child components via props.



** App is the single data source; data flows down to children via props.*

5. Data Flow

Step	Layer	Description
1	MongoDB Atlas	MauiBnB database stores the 'properties' collection with 5 seeded Maui B&B records
2	Express / Mongoose	GET /properties/:id retrieves one property document and returns it as JSON; CORS enabled
3	React useEffect	App component calls the API endpoint when the page first mounts
4	React useState	API response stored in state; loading and error states handled separately
5	Child Components	App passes name, description, amenities as props; children render the data

6. Acceptance Criteria

AC-1: Given the Express server is running, when the marketing page loads, then all four sections are visible within 3 seconds.

AC-2: Given a 375px mobile viewport, when the page renders, then all content is readable without horizontal scrolling.

AC-3: Given a visitor clicks 'Reserve Your Stay', when clicked, then the default email client opens with the contact address pre-filled.

AC-4: Given the API returns property data, when Amenities renders, then every amenity from MongoDB appears as a card.

AC-5: Given the Express server is offline, when the page loads, then a clear error message is shown instead of a blank page.

7. Plans for Weeks 13–15

Week 13 — Visitor Dashboard (React Advanced):

Build a React Dashboard with three Chart.js visualizations: (1) bar chart of Maui visitor arrivals by month from DBEDT data, (2) donut chart of domestic vs. international visitors, (3) metric cards for avg. daily rate and occupancy. Add OpenWeatherMap widget for Maui weather. Add island dropdown filter.

Week 14 — Admin Authentication:

Implement Passport.js local strategy login. Create /admin/login and /admin/dashboard routes. Hash passwords with bcrypt (10+ salt rounds). Add CSRF protection. Build admin feature to post special offers visible on the public marketing page.

Week 15 — Hardening, Testing & Deployment:

Add Helmet.js security headers. Validate all form inputs with express-validator. Write 3+ Jest + Supertest integration tests. Complete security checklist (no hardcoded secrets, no injection vulnerabilities). Deploy to Render free tier and commit live URL to README.

AI Attribution: This PRD was drafted with assistance from Claude (Anthropic). All content reflects the student's own project decisions and plans.